RAMCO INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ELECOM MAGAZINSIO - TECHNICAL MAGAZINE

2019-2020 (ODD)

Volume 1 Issue 1

About the Department

The Department of Electronics and Communication Engineering (ECE) is established in the year 2013 with a vision to Develop futuristic technologist in Electronics and Communication Engineering to meet the global challenges and the societal needs of our nation. At present RIT offers a 4-year B.E. ECE programme with a sanctioned intake of 120 students. The Laboratories are equipped with latest technology in the fields of Electronic Devices and Circuits, Digital Electronics, Linear Integrated Circuits, Microprocessors & Microcontrollers, Very Large Scale Integration Circuits, Digital Signal Processing, Communication System, Computer Networks, Optical & Microwave, Embedded Systems that offer opportunities on a wide range of hardware and advanced software packages. In addition to this, the Department has equipped with Research Laboratories such as NI-LabVIEW Research Laboratory, Tessolve Semiconductor Laboratory and Technical Incubation Centre to enrich the knowledge of students and faculty.

The Department encompasses professional associations such as Students Technical Association "Electronica", IETE, ISTE chapters which provides a platform to learn professional and soft skills. The Department had organized various Value Added Courses, Electronica-Association activities, Workshops, Conference, Guest lectures and Industrial Visits. The students had excelled in both co-curricular and extra curricular activities. The faculty members also contributed their efforts to get sponsorship from FAER, TNSCST etc. The Department delights to release the Technical Magazine - **"ELECOM MAGAZINSIO"** for the academic year 2019-2020 Odd Semester.

RAMCO INSTITUTE OF TECHNOLOGY

VISION

To evolve as an Institute of international repute in offering high-quality technical education, Research and extension programmes in order to create knowledgeable, professionally competent and skilled Engineers and Technologists capable of working in multi-disciplinary environment to cater to the societal needs.

MISSION

To accomplish its unique vision, the Institute has a far-reaching mission that aims:

- To offer higher education in Engineering and Technology with highest level of quality, professionalism and ethical standards
- To equip the students with up-to-date knowledge in cutting-edge technologies, wisdom, creativity and passion for innovation, and life-long learning skills
- To constantly motivate and involve the students and faculty members in the education process for continuously improving their performance to achieve excellence.

QUALITY POLICY

RIT aims to impart Quality Education in Engineering and Technology through an effective teaching-learning process, up-gradation of facilities and human resources, collaborating with Industry for promoting training and placement, research and consultancy activities with a commitment to continual improvement of Quality Management System.

ELECTRONICS AND COMMUNICATION ENGINEERING

VISION & MISSION

PEOs & PSOs

VISION

Develop futuristic technologist in Electronics and Communication Engineering to meet the global challenges and the societal needs of our nation.

MISSION

- Educate students to become enlightened engineers with the latest know-how to take on the challenges concerned with the society.
- Equip the students with the current trends and latest technologies in the field of Electronics & Communication Engineering to make them technically strong and ethically sound.
- Constantly motivate the students and faculty members to achieve excellence in Electronics & Communication Engineering and Research & Development activities.

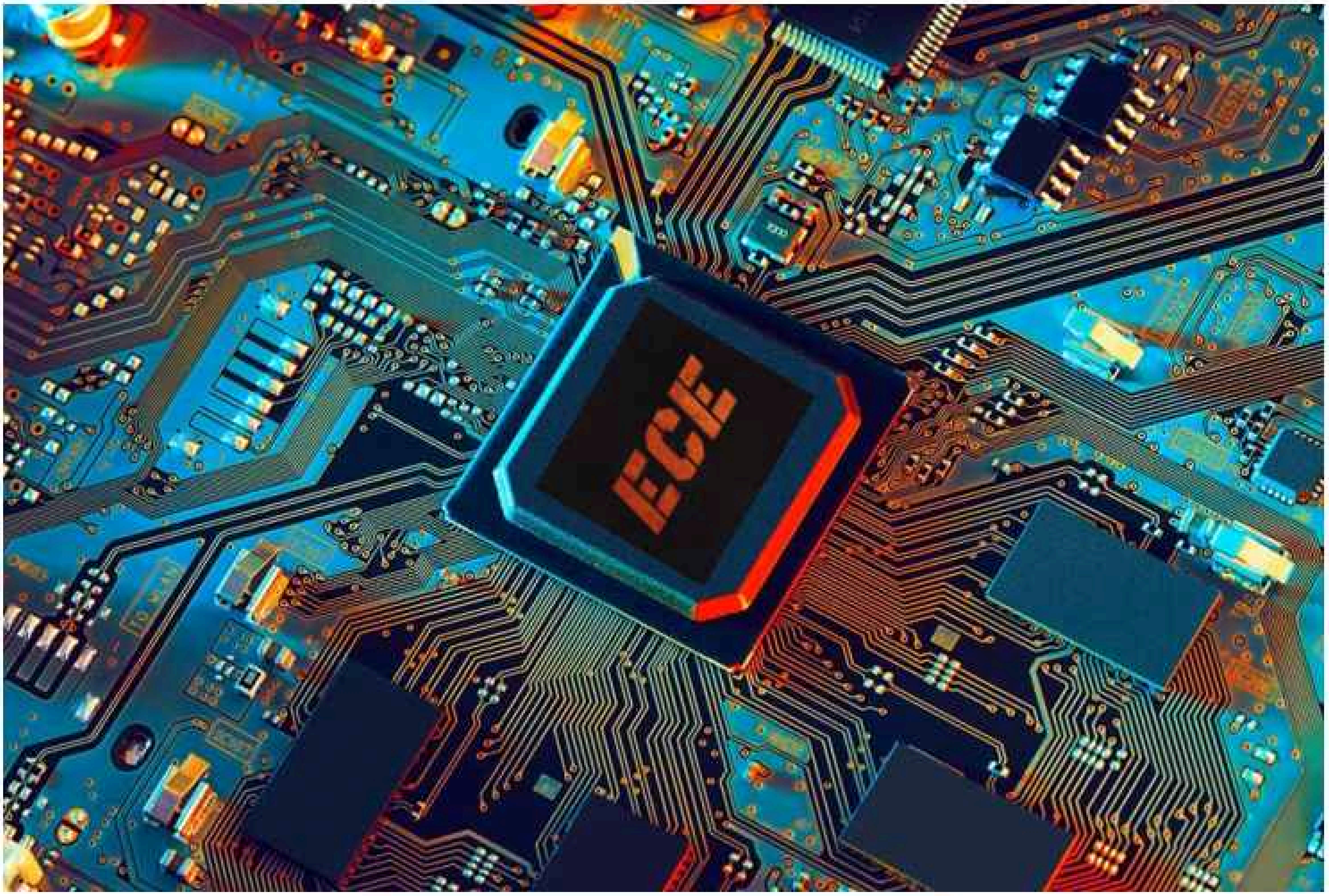
PPROGRAM EDUCATIONAL OBJECTIVES (PEOs) :

- Graduates will gain knowledge through continuous learning to enhance competency and career prospects.
- Graduates will be technically competent to design, analyze, develop and implement Electronic and Communication Systems.
- Graduates will apply the basic fundamental concepts in Electronic and related fields to solve general engineering Problems related to societal needs.
- Graduates will be professionals with attributes, passion, positive attitude, ethics and moral values for a successful career.
- Graduates will Communicate effectively and interact responsibly with colleagues, employees, team members, clients and the society.

PROGRAM SPECIFIC OUTCOMES (PSOs) :

After successful completion of the degree, the students will be able to:

- Adapt to emerging trends in Information and Communication Technology by innovating new ideas to solve existing/novel problems.
- Excel in latest technologies of Embedded Systems.
- Design, analyze and develop electronic products in the area of VLSI Design and Communication Systems.
- Contribute as Entrepreneurs in building electronic products.



INSIDE THIS ISSUE

Engineering Services Examination

The **Indian Engineering Services** (IES) are the services which cater to the technical and managerial functions of the Government of India in the field of engineering. As in most countries, the Indian government recruits its civil servants and officials through competitive examinations. Many candidates take the exams, competing for a limited number of posts. IES officers are selected by the union government on the recommendation of the Union Public Service Commission (UPSC).

General Studies and Engineering Aptitude (Preliminary)

1. Current issues of national and international importance relating to social, economic and industrial development.
2. Engineering Aptitude covering Logical reasoning and Analytical ability.
3. Engineering Mathematics and Numerical Analysis.
4. General Principles of Design, Drawing, Importance of Safety.
5. Standards and Quality practices in production, construction, maintenance and services.
6. Basics of Energy and Environment: Conservation, environmental pollution and degradation, Climate Change, Environmental impact assessment.
7. Basics of Project Management.
8. Basics of Material Science and Engineering. Information and Communication Technologies (ICT) based tools and their applications in Engineering such as networking, e-governance and technology based education.
9. Ethics and values in Engineering profession.

PAPER – I

1. Basic Electronics Engineering .
2. Basic Electrical Engineering .
3. Materials Science .
4. Electronic Measurements and Instrumentation .
5. Network Theory .
6. Analog and Digital Circuits.

PAPER – II

1. Analog and Digital Communication Systems
2. Control Systems
3. Computer Organization and Architecture
4. Electro Magnetics
5. Advanced Electronics Topics
6. Advanced Communication Topics

By

S.Bhuvan Akshay

III Year ECE-A

Haptic Technology

Haptic technology, also known as kinesthetic communication or 3D touch, refers to any technology that can create an experience of touch by applying forces, vibrations, or motions to the user.

Used to create virtual objects in a computer simulation, to control virtual objects , and to enhance remote control of machines and devices

Advantages of Haptic Technology :

Digital world can be experienced and perceived. Easily accessible and user friendly. Accuracy and precision is high.

Disadvantages of Haptic Technology:

- Involves complex designing as Haptic devices requires precision of touch.
- High initial cost involved

Haptic devices may incorporate tactile sensors that measure forces exerted by the user on the interface.

Simple haptic devices are common in the form of game controllers, joysticks, and steering wheels

Haptic technology facilitates investigation of how the human sense of touch works by allowing the creation of controlled haptic virtual objects.

Most researchers distinguish three sensory systems related to sense of touch in humans: cutaneous, kinesthetic and haptic

Haptic Technology

Uses:

- Haptic refers to technology that uses touch to control and interact with computers. A user may apply a sense of touch through vibrations, motion or force.
- Haptic technology is used mainly in creating virtual objects, controlling virtual objects or in the improvement of the remote control of machines and devices.
- Haptic technology is also widely used in teleportation , or tele-robotics. In a telerobotic system, a human operator controls the movements of a robot that is located some distance away. Some telerobots are limited to very simple tasks, such as aiming a camera and sending back visual images.

Applications:

- Helping the blind to feel city
- Tactile touchscreen in Nokia mobile phones

By
R.Anushya
III Year ECE-A

GOOGLE SELLS OUR FUTURE

Google is the dominant search engine in the world with billions of page views. Google Adwords are extremely powerful when it comes to targeting. Businesses can find out the exact keywords that people type when they are looking for a certain product or service. That's how they can target this people who have interest for purchasing. As long as you will go to google.com to search for stuff on it, you can see the same stuff from different websites.

Those websites are the partners of Google and that display ads on their pages. Let us take an example. You visit a website to buy a camera. This website will track your visit by installing a cookie on your computer. Later, when you might go to another website, you might see the ad from that previous camera website somewhere on it .

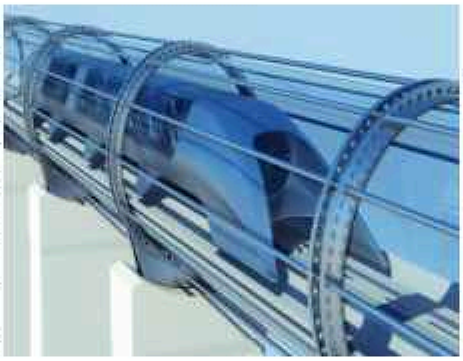
HOW DOES THIS HAPPEN?

A company pays Google to show a specific visual ads to all the people who visited its website on the other websites of Google Adsense Network and also it is possible to target this website's theme or even by domain name. That's how Google is selling our data to many other websites and make money from it.

By
S.Isha Suga Nandhini
III Year ECE-A

HYPERLOOP

Hyperloop is a concept for very high speed, fixed guide way, intercity surface transportation, using capsule like vehicles that operate in sealed partial vacuum tubes. At this stage, the technology is unproven, but it has elicited a great deal of interest from journalists, Investors, engineering firms, and governments. This research is being conducted to provide a high level evaluation of hyperloop in terms of its commercial potential, environmental impact, costs, safety issues, and regulatory and policy issues and to identify further research topics related to the technology. This research is intended to provide NASA decision makers with appropriate context to make decisions on the future direction of NASA's involvement in Hyperloop research.



A Hyperloop is a proposed mode of passenger and freight transportation. It is first used to describe an open-source vactrain design released by a joint team from Tesla and SpaceX. Hyperloop is a sealed tube or system of tubes through which a pod may travel free of air resistance or friction conveying people or objects at high speed while being very efficient, thereby drastically reducing travel times over medium-range distances. Elon Musk's version of the concept, first publicly mentioned in 2012, incorporates reduced-pressure tubes in which pressurized capsules ride on air bearings driven by linear induction motors and axial compressors.

The Hyperloop Alpha concept was first published in August 2013, proposing and examining a route running from the Los Angeles region to the San Francisco Bay Area, roughly following the Interstate 5 corridor. The Hyperloop Genesis paper conceived of a hyperloop system that would propel passengers along the 350-mile (560 km) route at a speed of 760 mph (1,200 km/h), allowing for a travel time of 35 minutes, which is considerably faster than current rail or air travel times. Hyperloop One announced Wednesday that it successfully tested a full hyperloop. The step into the future occurred in May at the company's Nevada test track, where engineers watched a magnetically levitating test sled fire through a tube in near-vacuum, reaching 70 mph in just over five seconds. The Hyperloop train could cut down the travel between Mumbai to Pune down to 23 minutes.

By
Libya Joseph
III Year ECE-A

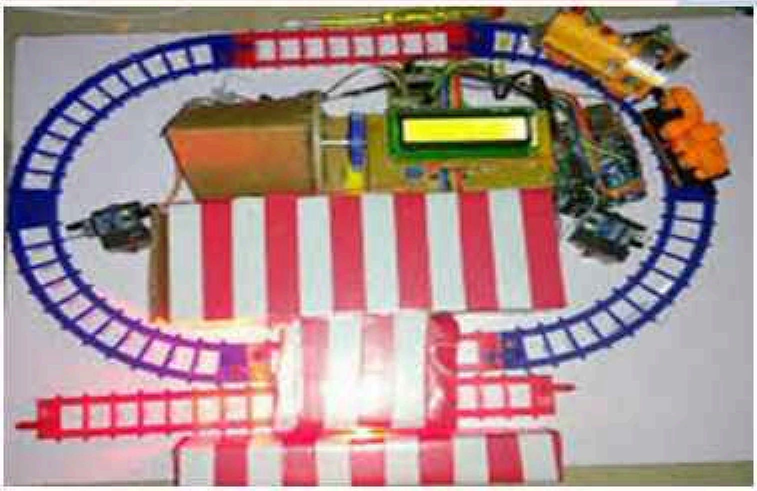
AUTOMATED FOOT OVER BRIDGE

Indian Railways was established on 16 April, 1853. For the past decades, the vertical lift forms, swings and foot over bridges were used in railway stations. But it is difficult for aged and handicapped persons to reach the other platform through the foot-over bridges. To provide the flexible bridge for the railway platforms in order to overcome these difficulties in the usage of foot-over bridges in railway stations. It is planned to go with a pair of IR (Proximity) sensors on either side of the railway station at certain distance. Initially, the flexible bridge is connected between the platforms which are fixed at one end and other end is movable by the functioning of the sensors and control unit.

* Arduino
* Servo motor
* IR sensor
* Led display

FUTURE EXTENT

In future we can reduce accidents by using the camera pixels. By extending this Project the Platform of each train can be easily detected by each other. Train can transmit the info to station when the train is within some distance range from station.



Done by
A.Manikandan
M.Muthukrishnan
V.Vasantha Kumar
M.Sankar
III Year ECE-B

Engineer's Day Project

அப்துல் கலாம்

அக்னி சிறகணிந்த ஏவுகணையாய்...
அணையாது எரியும் ஆசிய ஜோதியாய்...
இளைய உள்ளங்களின் கனவுகளாய்...
என்றும் நிற்கும் எம் நாயகனுக்கு
காற்றும் சொல்லும் சலாம் என்று!!!

அம்மா

சிசுவான எனக்காக
சித்ரவதை அனுபவித்து
பசியின்றியும் எனக்காக
பசிதீர உணவளித்து
உதிரத்தை உணவாக்கி
உதைகளையும் சுகமாக்கி
ஆசையாய் எனையே
அழகாக உருவாக்கி
ஆனந்தம் கண்ட
தெய்வம் அவள்...

அகிலத்தின் முதல் அடியை
உன் வயிற்றில் வைத்தேன்

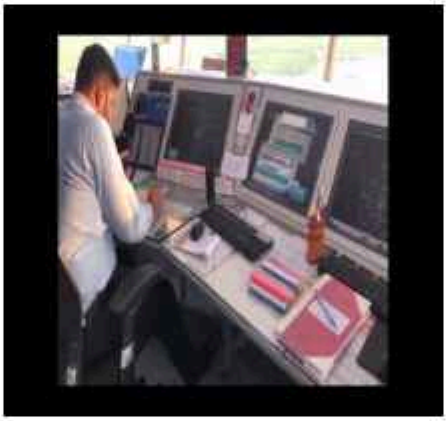
அவை தழும்பாயினும்
வலிதந்த எனக்கு
அவள் தந்த தண்டனை
ஆசை முத்தம்...
உதிர்த்தேன் முதல் துளி கண்ணீரையும் உனக்காய்!

மு.ஆயிஷா ஷஹானா
மூன்றாம் ஆண்டு

COMMUNICATION NAVIGATION AND SURVEILLANCE IN AIRPORTS

INTRODUCTION TO AAI

The important function of AAI and is to maintain and manage Civil Aviation Infrastructure both on ground and airspace in the country by providing communication and Navigation aids .



COMMUNICATION

In airport, communication is established between pilot to controller using very high frequency and high frequency range .VHF have high fidelity and clarity makes easier process

NAVIGATION

The introduction about the navigation and its equipment's are discussed. The following equipment's are used to provide accurate and reliable information .

- ILS
- DVOR
- NDB

SURVEILLANCE

An airport surveillance radar (ASR) is a radar system used at airports to detect and display the presence and position of aircraft in the terminal area, the airspace around airports. It is the main air traffic control system for the airspace around airports.

AIR TRAFFIC MANAGEMENT SERVICES

ATMS are done by air traffic control officers .They regulate the traffic in the Indian Airspace .The main objective is to prevent collisions between aircraft ,provide safe orderly flow of traffic and to efficiently reduce delays.

COMMUNICATION NAVIGATION AND SURVEILLANCE IN AIRPORTS

NDB-Non directional Beacon .

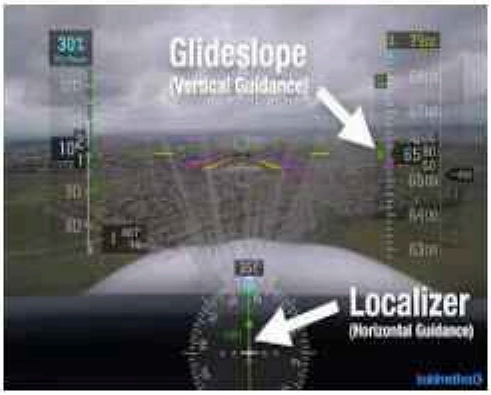
NDB is an Transmitter antenna used for offshore platforms received by Automatic Direction Finder (ADF) in on board aircraft

DVOR –DOPPLER VERY HIGH OMNI RANGE FREQUENCY

DVOR provides information to aircraft to define air traffic control routes for reroute .

INSTRUMENT LANDING SYSTEM

ILSis also called as Extended electronic runway used for landing approach .ILS consists of Localizer, Glide path ,Distance measuring equipment.

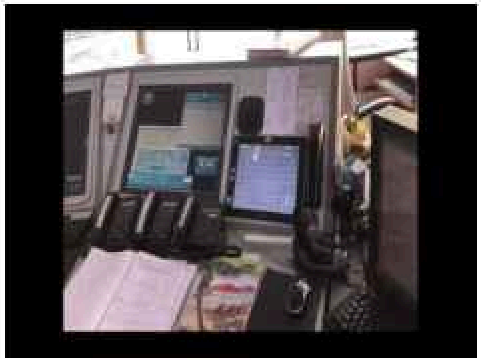


Localizer – It is used to identify the central line of runway to align the aircraft for efficient landing.

Glide path – It is used to land at an angle 3 degree for safe angle of movement .

DME – It is used to measure the slant distance from airport to aircraft.

DATIS – Digital Automatic Terminal Information System.



DATIS is used for providing METROLOGICAL INFORMATION such as Wind direction, visibility, Temperature, Humidity, Local air pressure, etc..

By
D.Joseph Chandrasekar
R.Arun Prasanth
III Year ECE-A

Your Happiness

Happiness is a feel which instinct your mind to concentrate on your own beauty. Here the beauty means the real character that comes from the heart.

Our happiness cannot be given by others. Only we are the responsible to find our own way of being happy always.

Two things can ruin our happiness, First, Living in the past.

Second thing is observing others.

If we abort these two things, no one can take away the happiness from us.

Always live happy with what you have. God has given a unique life style for us to live, don't try to copy other's life style.

But one thing, never kill others happiness for the sake of our happiness. To be happy, spread love and positivity as much as possible, no matter how much others hurt us, just keep smiling and move on.

Many times, exaggerating on something and overthinking destroys one's happiness. So strengthen your mind to take things as it is, life will be even more happy.

We want to live by each other's happiness not by Each other's Misery - Charlie Chaplin

HAPPINESS

IS

THE

BEST

MAKEUP

HapPiness Initiative

By
Shrinithimeena .M
I Year ECE-B

Reorganizing a computer chip : Transistors can now both process and store information

Date: December 9, 2019
Source: Purdue University

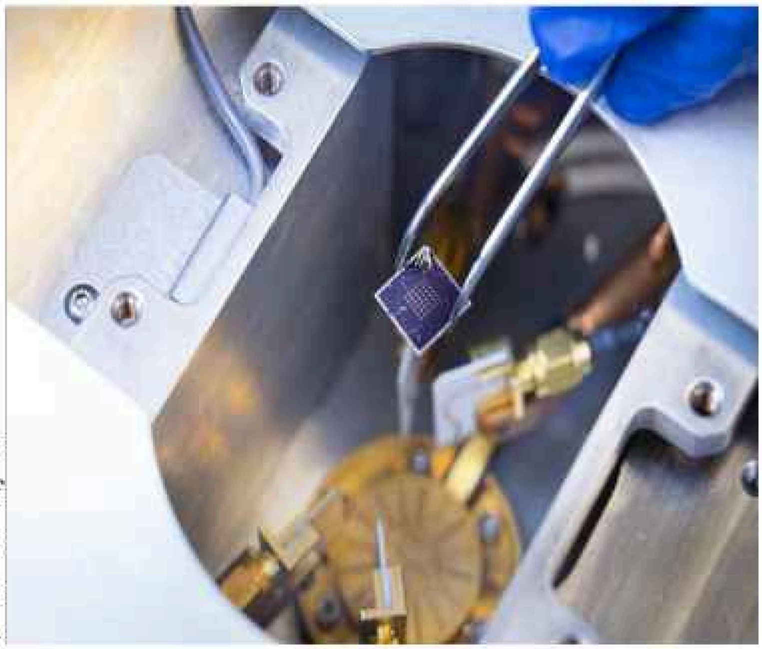
Researchers solve decades-old challenge of building a functional transistor integrated with ferroelectric RAM

Summary

Researchers have created a more feasible way to combine transistors and memory on a chip, potentially bringing faster computing.

Full Article

A computer chip processes and stores information using two different devices. If engineers could combine these devices into one or put them next to each other, then there would be more space on a chip, making it faster and more powerful.



Purdue University engineers have developed a way that the millions of tiny switches used to process information -- called transistors - could also store that information as one device.

The method, detailed in a paper published in *Nature Electronics*, accomplishes this by solving another problem: combining a transistor with higher-performing memory technology than is used in most computers, called ferroelectric RAM

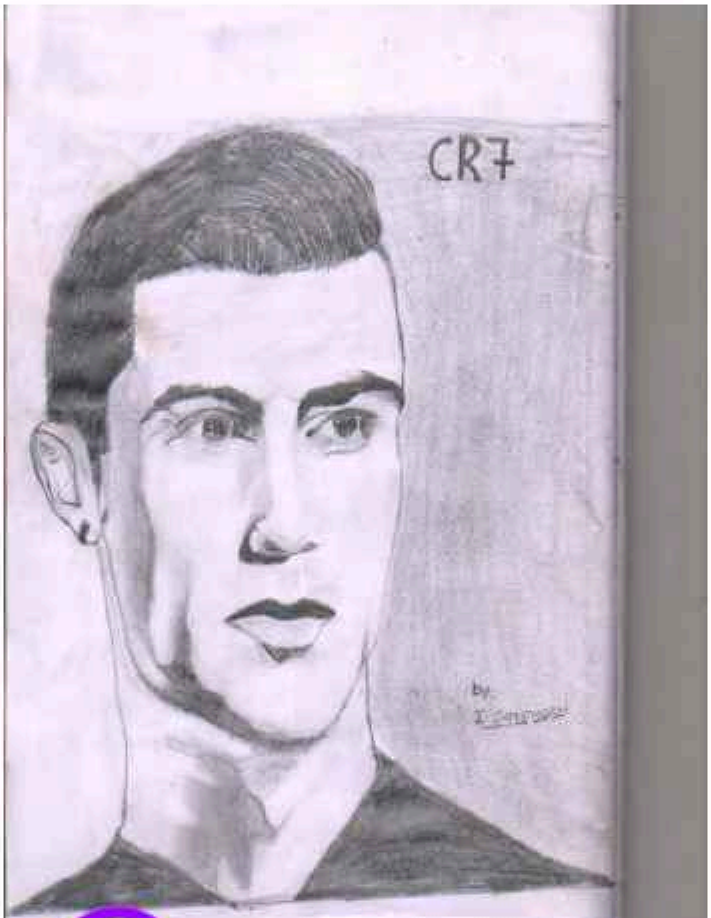
Researchers have been trying for decades to integrate the two, but issues happen at the interface between a ferroelectric material and silicon, the semiconductor material that makes up transistors. Instead, ferroelectric RAM operates as a separate unit on-chip, limiting its potential to make computing much more efficient.

A team led by Peide Ye, the Richard J. and Mary Jo Schwartz Professor of Electrical and Computer Engineering at Purdue, discovered how to overcome the mortal enemy relationship between silicon and a ferroelectric material

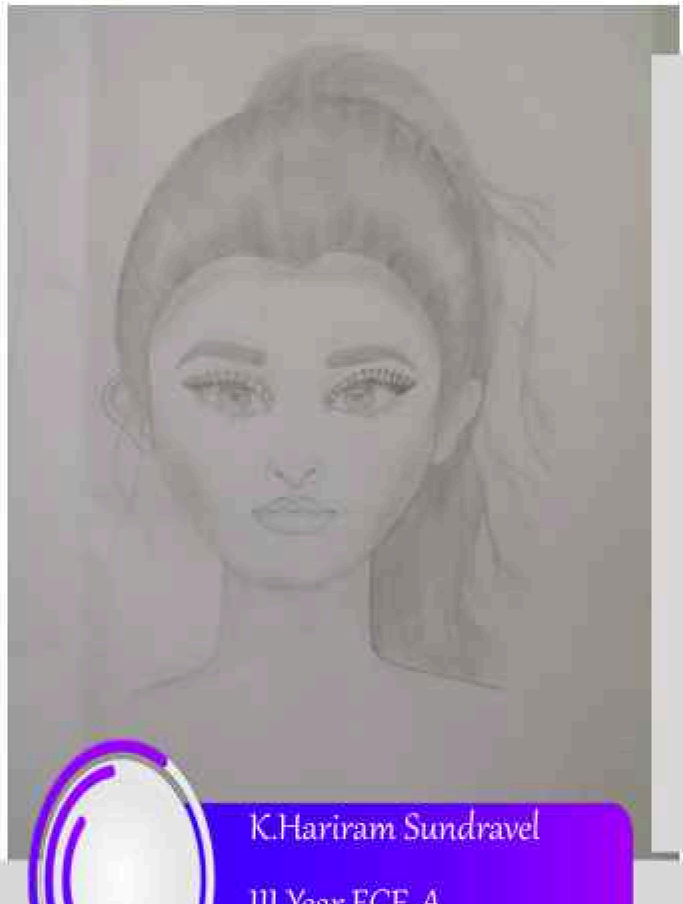
This research was performed in the Purdue Discovery Park Brick Nanotechnology Centre and supported by the National Science Foundation, Air Force Office of Scientific Research, Semiconductor Research Corporation, Defence Advanced Research Projects Agency and the U.S. Office of Naval Research

By
M.MuthuKrishnan
A.Manikandan
III year ECE-B

A.Sushma , I Year ECE-B



I.Santhosh , I Year ECE-B



K.Hariram Sundravel

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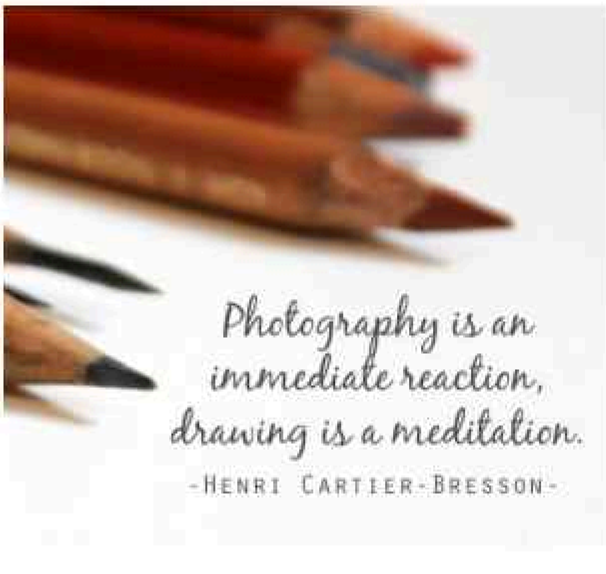

Lahvanya, III Year ECE-A

PC : M.Aniruth Shriram

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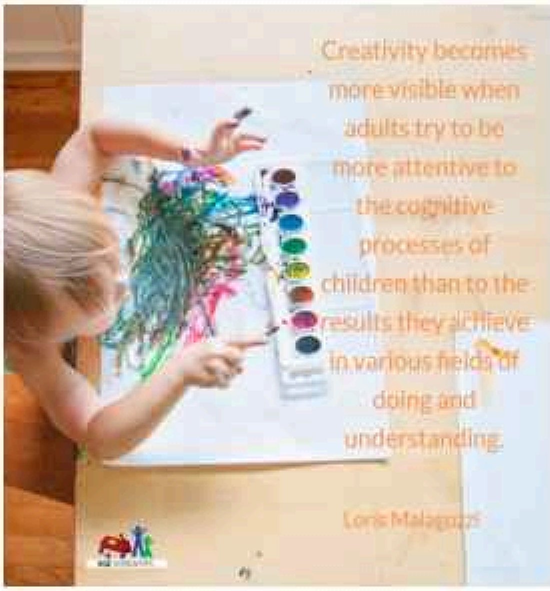
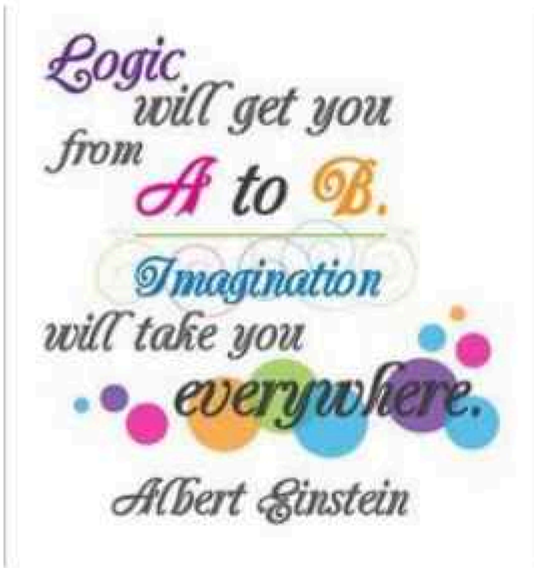

Creativity becomes more visible when adults try to be more attentive to the cognitive processes of children than to the results they achieve in various fields of doing and understanding.
Loris Malaguzzi


Logic will get you from A to B.
Imagination will take you everywhere.
Albert Einstein


"Talent wins games, but teamwork and intelligence wins championships."
- Michael Jordan

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